MCS 320 Quiz 10 Friday 6 November 2026

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. For n variables $x_1, x_2, \ldots, x_n$, define the matrix

$$S_n = \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_n \\ x_2 & x_1 & x_2 & \cdots & x_{n-1} \\ x_3 & x_2 & x_1 & \cdots & x_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n & x_{n-1} & x_{n-2} & \cdots & x_1 \end{bmatrix}.$$

   Apply your definition for $n = 8$ and compute S_8.

2. Let $x(t)$, a function of t, be the solution of the initial value problem

$$\frac{d^2}{dt^2}x(t) - 2\frac{d}{dt}x(t) - 3x(t) = 0, \quad x(0) = 4, \quad x'(0) = 0.$$

   Compute $x(t)$ using the Laplace transform.

   What is the algebraic equation that must be solved?